

# TestPlatform

## Rapport d'Execution de Campagne

**test**

Projet: ms\_gestion

Version: 2.5

Environnement: ms\_gestion\_env

Debut: 29/06/2026 17:03 | Fin: 29/06/2026 17:05

Duree: 125.8 s

Rapport genere le 29/06/2026 17:05

**PARTIEL**

## Resume executif

| Total | Succes | Echecs | Erreurs | Duree   |
|-------|--------|--------|---------|---------|
| 3     | 2      | 1      | 0       | 125.8 s |

### Taux de reussite



Evolution vs campagne precedente: N/A

Premiere campagne

**Corrections IA: 2 test(s) corrige(s) automatiquement, 2 avec succes**

## Informations du contexte

|             |            |
|-------------|------------|
| Projet      | ms_gestion |
| Description | test       |

## Suites

| Nom               | Type        | Depot Git | Branche |
|-------------------|-------------|-----------|---------|
| Suite UNIT        | UNIT        | N/A       | main    |
| Suite INTEGRATION | INTEGRATION | N/A       | main    |

## Cas de test

| Titre              | Type        | Priorite | Risque   | Script |
|--------------------|-------------|----------|----------|--------|
| create project     | UNIT        | 2        | HIGH     | IA     |
| create envirenemnt | INTEGRATION | 5        | CRITICAL | IA     |
| List environnement | INTEGRATION | 2        | CRITICAL | IA     |

## Environnement

|                 |                                                                                    |
|-----------------|------------------------------------------------------------------------------------|
| URL Web         | http://localhost:3000                                                              |
| URL API         | N/A                                                                                |
| Repository      | https://github.com/ilyeschaabani/Automatisation_intelligente_des_tests_applicatifs |
| Branche         | main                                                                               |
| Base de donnees | POSTGRESQL                                                                         |

# Resultats details

## Synthese par suite

| Suite             | Total | Succes | Echecs | Taux |
|-------------------|-------|--------|--------|------|
| Suite UNIT        | 1     | 1      | 0      | 100% |
| Suite INTEGRATION | 2     | 1      | 1      | 50%  |

## Suite UNIT

|                |      |        |                       |         |
|----------------|------|--------|-----------------------|---------|
| create project |      | IA x1  |                       | SUCCESS |
| Type           | UNIT | Duree  | 11.5 s                |         |
| Script         | IA   | Erreur | [COMPILATION_ERROR] — |         |

## Suite INTEGRATION

|                    |             |        |                       |         |
|--------------------|-------------|--------|-----------------------|---------|
| create envirenemnt |             | IA x1  |                       | SUCCESS |
| Type               | INTEGRATION | Duree  | 11.0 s                |         |
| Script             | IA          | Erreur | [COMPILATION_ERROR] — |         |

|                    |             |        |                                                                                                                                                        |
|--------------------|-------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| List environnement |             |        | FAILURE                                                                                                                                                |
| Type               | INTEGRATION | Duree  | 25.1 s                                                                                                                                                 |
| Script             | IA          | Erreur | [COMPILATION_ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.95 s <<< FAILURE! -- in suites.integration.EnvironmentServiceIT |

# Analyse IA

## create project

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Erreurs<br>extraites | <pre>[INFO] Scanning for projects... [INFO] [INFO] -----&lt; com.pfe.platform:ms_gestion &gt;----- [INFO] Building ms_gestion 0.0.1-SNAPSHOT [INFO]    from pom.xml [INFO] -----[ jar ]----- [INFO] [INFO] --- resources:3.3.1:resources (default-resources) @ ms_gestion --- [INFO] Copying 1 resource from src\main\resources to target\classes [INFO] Copying 0 resource from src\main\resources to target\classes [INFO] [INFO] --- compiler:3.14.1:compile (default-compile) @ ms_gestion --- [INFO] Nothing to compile - all classes are up to date. [INFO] [INFO] --- resources:3.3.1:testResources (default-testResources) @ ms_gestion --- [INFO] skip non existing resourceDirectory C:\tmp\ms-execution\exec- 3421550725211656858\Backend\ms_gestion\src\test\resources [INFO] [INFO] --- compiler:3.14.1:testCompile (default-testCompile) @ ms_gestion --- [INFO] Recompiling the module because of changed source code. [INFO] Compiling 2 source files with javac [debug parameters release 17] to target\test-classes [INFO] [INFO] --- surefire:3.2.5:test (default-test) @ ms_gestion --- [INFO] Using configured provider org.apache.maven.surefire.testng.TestNGProvider [INFO] [INFO] ----- [INFO]  T E S T S [INFO] ----- [INFO] Running suites.unit.ProjectServiceCreateHappyTest [INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.127 s -- in suites.unit.ProjectServiceCreateHappyTest [INFO] [INFO] Results: [INFO] [INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0 [INFO] [INFO] ----- [INFO] BUILD SUCCESS [INFO] ----- [INFO] Total time:  7.497 s [INFO] Finished at: 2026-06-29T14:35:36+01:00 [INFO] -----  ----- STDERR ----- Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended</pre> |
| Analyse<br>IA        | <pre>### 1. Ce qui est bien testé Le test couvre la logique *happy path* de la création de projet : validation de l'unicité du nom, sauvegarde du projet, création de l'enregistrement utilisateur comme membre, et retour des bonnes valeurs dans la réponse. L'utilisation de `MockedStatic` pour `SecurityUtils.getCurrentUserId()` et la simulation correcte des dépendances (`projectRepository`, `projectMemberRepository`) garantissent un contrôle précis de l'environnement de test.  ### 2. Suggestion d'amélioration</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Le test manque de \*\*vérification de l'ordre ou de la cohérence entre les appels\*\* (ex. : garantir que le projet est sauvegardé \*avant\* la création du membre). De plus, les assertions sur la réponse pourraient inclure `assertEquals(result.getId(), 1L)` pour valider que l'ID est bien renseigné, améliorant ainsi la robustesse. Un petit bémol : on pourrait extraire la configuration des mocks (`when(...).thenReturn(...)`) dans une méthode helper pour une lisibilité accrue.

## create envirenemnt

**Erreurs extraites**

```
Hibernate: drop table if exists environments cascade
Hibernate: drop table if exists project_members cascade
Hibernate: drop table if exists projects cascade
Hibernate: drop table if exists security_scans cascade
Hibernate: drop table if exists security_vulnerabilities cascade
Hibernate: drop table if exists test_cases cascade
Hibernate: drop table if exists test_suites cascade
Hibernate: create table campaign_testcases (execution_order integer, campaign_id bigint not null, test_case_id bigint not null, primary key (campaign_id, test_case_id))
Hibernate: create table campaigns (created_at timestamp(6), environment_id bigint not null, finished_at timestamp(6), id bigint generated by default as identity, project_id bigint not null, started_at timestamp(6), app_version varchar(255), git_branch varchar(255), name varchar(255) not null, status varchar(255) check ((status in ('PENDING','RUNNING','FINISHED','FINISHED_WITH_ERRORS','ABORTED'))), trigger_mode varchar(255) check ((trigger_mode in ('MANUAL','SCHEDULED','CI'))), primary key (id))
Hibernate: create table compliance_results (failed_controls integer, passed_controls integer, score float(53), total_controls integer, evaluated_at timestamp(6), id bigint generated by default as identity, project_id bigint, framework varchar(255) not null, version varchar(255), primary key (id))
Hibernate: create table environments (created_at timestamp(6), id bigint generated by default as identity, project_id bigint not null, base_url_api varchar(255), base_url_web varchar(255), database_type varchar(255), git_branch varchar(255), git_repo_url varchar(255), name varchar(255) not null, primary key (id))
Hibernate: create table project_members (id bigint generated by default as identity, project_id bigint not null, user_id bigint not null, primary key (id), unique (project_id, user_id))
Hibernate: create table projects (created_at timestamp(6), created_by bigint, id bigint generated by default as identity, description TEXT, git_default_branch varchar(255), git_repo_url varchar(255), name varchar(255) not null unique, status varchar(255) check ((status in ('ACTIVE','PAUSED','ARCHIVED'))), primary key (id))
Hibernate: create table security_scans (coverage float(53), files_analyzed integer, lines_analyzed integer, completed_at timestamp(6), id bigint generated by default as identity, project_id bigint, started_at timestamp(6), branch varchar(255), commit_hash varchar(255), duration varchar(255), engine varchar(255) not null, scan_ref varchar(255) not null, scan_type varchar(255) not null check ((scan_type in ('SAST','DAST','SCA'))), status varchar(255) not null check ((status in ('RUNNING','COMPLETED','FAILED','SCHEDULED'))), target_url varchar(255), primary key (id))
Hibernate: create table security_vulnerabilities (line integer, created_at timestamp(6), id bigint generated by default as identity, project_id bigint, scan_id bigint, updated_at timestamp(6), assigned_to varchar(255), cwe_id varchar(255), description TEXT, endpoint varchar(255), evidence TEXT, file varchar(255), fix_example TEXT, http_method varchar(255), owasp_category varchar(255), parameter varchar(255), recommendation TEXT, risk TEXT, severity varchar(255) not null check ((severity in ('CRITICAL','HIGH','MEDIUM','LOW','INFO'))), snippet TEXT, source varchar(255), status varchar(255) not null check ((status in ('OPEN','IN_PROGRESS','RESOLVED','CLOSED','FALSE_POSITIVE'))), title varchar(255) not null, vuln_type varchar(255) not null check ((vuln_type in ('SAST','DAST','SCA'))), primary key (id))
Hibernate: create table test_cases (active boolean, flaky boolean, generated boolean not null, max_duration_seconds integer, priority integer, created_at timestamp(6), id bigint generated by default as identity, suite_id bigint not null, database_type varchar(255), description TEXT, generated_code TEXT, git_repo_url varchar(255), risk_level varchar(255) check ((risk_level in ('CRITICAL','HIGH','MEDIUM','LOW'))), script_path varchar(255),
```

```

spring_profile varchar(255), tags varchar(255), target_class_name varchar(255), title
varchar(255) not null, type varchar(255) check ((type in
('WEB','API','UNIT','INTEGRATION','UX_WEB','UX_MOBILE'))), test_data JSONB, primary key
(id))
Hibernate: create table test_suites (created_at timestamp(6), id bigint generated by default
as identity, project_id bigint not null, description TEXT, git_branch varchar(255),
git_repo_url varchar(255), module_path varchar(255), name varchar(255) not null, type
varchar(255) check ((type in ('WEB','API','UNIT','INTEGRATION','UX_WEB','UX_MOBILE'))),
primary key (id))
Hibernate: alter table if exists campaign_testcases add constraint
FKdyri4dtm81vciauhdyekpeh1t foreign key (campaign_id) references campaigns
Hibernate: alter table if exists campaign_testcases add constraint
FK9weg2djl986e8fnmb11onvyg6 foreign key (test_case_id) references test_cases
Hibernate: alter table if exists campaigns add constraint FKmyimryhxtshr5smm4vb6y7ny8
foreign key (environment_id) references environments
Hibernate: alter table if exists campaigns add constraint FK8tqxs91vmi30cb4pkapa432mi
foreign key (project_id) references projects
Hibernate: alter table if exists compliance_results add constraint
FKbc45lcknv6hj2rrknhhtt0rsu foreign key (project_id) references projects
Hibernate: alter table if exists environments add constraint FKnc1c5pk4efot15s7hyrqah6ru4
foreign key (project_id) references projects
Hibernate: alter table if exists project_members add constraint FKdki1sp2homqsdcvqm9yrix31g
foreign key (project_id) references projects
Hibernate: alter table if exists security_scans add constraint FKal1fccjhlqwsdiyfe0xgn8hag
foreign key (project_id) references projects
Hibernate: alter table if exists security_vulnerabilities add constraint
FK5dqkgk4x5b5h2qub892el4l45 foreign key (project_id) references projects
Hibernate: alter table if exists security_vulnerabilities add constraint
FK41kves1citph1miw1j7n7v8gi foreign key (scan_id) references security_scans
Hibernate: alter table if exists test_cases add constraint FKeuc59ejswhi3pn4lv997yd82
foreign key (suite_id) references test_suites
Hibernate: alter table if exists test_suites add constraint FK9m60ulvcghbrs28iuc98ug1vb
foreign key (project_id) references projects
2026-06-29T17:02:04.714+01:00 INFO 36476 --- [ms_gestion] [          main]
j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for
persistence unit 'default'
2026-06-29T17:02:04.873+01:00 INFO 36476 --- [ms_gestion] [          main]
o.s.d.j.r.query.QueryEnhancerFactories   : Hibernate is in classpath; If applicable, HQL
parser will be used.
2026-06-29T17:02:06.471+01:00 WARN 36476 --- [ms_gestion] [          main]
JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default.
Therefore, database queries may be performed during view rendering. Explicitly configure
spring.jpa.open-in-view to disable this warning
2026-06-29T17:02:06.617+01:00 WARN 36476 --- [ms_gestion] [          main]
.s.a.UserDetailsServiceAutoConfiguration :

Using generated security password: d7b440be-d2f1-4671-a95e-e351a9d29b45

This generated password is for development use only. Your security configuration must be
updated before running your application in production.

2026-06-29T17:02:06.630+01:00 INFO 36476 --- [ms_gestion] [          main]
r$InitializeUserDetailsManagerConfigurer : Global AuthenticationManager configured with
UserDetailsService bean with name inMemoryUserDetailsManager
2026-06-29T17:02:07.533+01:00 INFO 36476 --- [ms_gestion] [          main]
s.integration.EnvironmentServiceIT       : Started EnvironmentServiceIT in 12.637 seconds
(process running for 13.746)
Hibernate: insert into projects
(created_at,created_by,description,git_default_branch,git_repo_url,name,status) values
(?,?,?,?,?,?,?)
Hibernate: insert into project_members (project_id,user_id) values (?,?)
Hibernate: select pm1_0.id from project_members pm1_0 where pm1_0.project_id=? and

```

```
pm1_0.user_id=? fetch first ? rows only
Hibernate: select e1_0.id from environments e1_0 where e1_0.project_id=? and e1_0.name=?
fetch first ? rows only
Hibernate: insert into environments
(base_url_api,base_url_web,created_at,database_type,git_branch,git_repo_url,name,project_id)
values (?, ?, ?, ?, ?, ?, ?, ?)
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 14.40 s -- in
suites.integration.EnvironmentServiceIT
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 20.658 s
[INFO] Finished at: 2026-06-29T17:02:09+01:00
[INFO] -----

----- STDERR -----
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes
because bootstrap classpath has been appended
```

Analyse  
IA

1. **Ce qui est bien testé** :

Le test couvre correctement la *happy path* de la méthode `add` d'`EnvironmentService`, incluant la validation des données d'entrée, la persistance en base (via `EnvironmentRepository`), la gestion du contexte sécurité (via `SecurityContextHolder`), et la vérification de l'appartenance à un projet (`ProjectMember`). L'environnement de test H2 en mémoire et la configuration `@Transactional + @Rollback` garantissent l'isolation et le nettoyage automatique.

2. **Suggestion d'amélioration** :

Ajouter des assertions sur la structure de l'objet retourné (ex : `@NotBlank` sur `name`, `@URL` sur les `baseUrl`) via un *validateur* (ex : `Validator.validate(request)`) pour s'assurer que les contraintes Bean Validation sont bien appliquées *avant* la persistance. Cela renforcerait la robustesse du test face à d'éventuels changements de contraintes métier.

## List environnement

Erreurs  
extraites

```
[ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.95 s <<<
FAILURE! -- in suites.integration.EnvironmentServiceIT
[ERROR] suites.integration.EnvironmentServiceIT.test_list_happy_path -- Time elapsed:
0.170 s <<< FAILURE!
[ERROR] Failures:
[ERROR] EnvironmentServiceIT>AbstractTestNGSpringContextTests.run:153-
>test_list_happy_path:78 Runtime Vous n'etes pas membre de ce projet
[ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-
plugin:3.2.5:test (default-test) on project ms_gestion: There are test failures.
[ERROR]
[ERROR] Please refer to C:\tmp\ms-execution\exec-
12038773517149811889\Backend\ms_gestion\target\surefire-reports for the individual test
results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump
and [date].dumpstream.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
```

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>[ERROR] For more information about the errors and possible solutions, please read the following articles:</p> <p>[ERROR] [Help 1] <a href="http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException">http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Analyse<br>IA | <p><b>**Cause**</b> : La méthode <code>environmentService.add(projectId, request)</code> échoue car le service vérifie l'appartenance à un projet via le contexte de sécurité, mais l'utilisateur <code>userId = 1L</code> n'est pas associé comme membre du projet sauvegardé (seul <code>createdBy</code> est fixé, sans rôle ou relation d'accès explicite).</p> <p><b>**Correction**</b> : Ajouter un utilisateur membre via un service de gestion des accès (ex. <code>projectAccessService.addMember(projectId, userId)</code>) <b>**ou**</b> configurer explicitement le <code>Project</code> avec une liste non vide de membres (ex. <code>project.setMembers(Collections.singleton(new ProjectMember(userId, Role.DEV)))</code>) <b>**avant**</b> le <code>save()</code>.</p> <p><b>**Conseil**</b> : Toujours s'assurer que le contexte de sécurité <b>**et**</b> les données relationnelles (ex. rôles, membres) correspondent à la logique métier cible <b>**avant**</b> d'exécuter les actions testées.</p> |

## Corrections automatiques par IA

L'IA a detecte des erreurs dans certains scripts generes et les a corriges automatiquement.

### create project

1 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-
3421550725211656858/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:[79,9]
cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-
3421550725211656858/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest
Résultat: Script corrigé (3412 chars), relance Maven...
Résultat: SUCCÈS après correction !
```

### create envirenemnt

1 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-
8096209451775046421/Backend/ms_gestion/src/test/java/suites/integration/EnvironmentServiceIT.java:[63,29]
incompatible types: com.pfe.platform.ms_gestion.entity.ProjectMemberId cannot be converted to java.lang.Long
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-80962
Résultat: Script corrigé (4760 chars), relance Maven...
Résultat: SUCCÈS après correction !
```

## Metriques

### Top 5 des tests les plus lents

| Test               | Suite             | Duree  |
|--------------------|-------------------|--------|
| List environnement | Suite INTEGRATION | 25.1 s |
| create project     | Suite UNIT        | 11.5 s |
| create envirenemnt | Suite INTEGRATION | 11.0 s |

### Temps d'execution par type

| Type        | Duree moyenne |
|-------------|---------------|
| UNIT        | 11.5 s        |
| INTEGRATION | 18.0 s        |

## Tests instables

Aucun test instable detecte

## Recommandations

- List environnement:

**\*\*Cause\*\*** : La méthode ``environmentService.add(projectId, request)`` échoue car le service vérifie l'appartenance à un projet via le contexte de sécurité, mais l'utilisateur ``userId = 1L`` n'est pas associé comme membre du projet sauvegardé (seul ``createdBy`` est fixé, sans rôle ou relation d'accès explicite).

**\*\*Correction\*\*** : Ajouter un utilisateur membre via un service de gestion des accès (ex.

``projectAccessService.addMember(projectId, userId)`` **\*\*ou\*\*** configurer explicitement le ``Project`` avec une liste non vide de membres (ex. ``project.setMembers(Collections.singleton(new ProjectMember(userId, Role.DEV)))``) **\*avant\*** le ``save()``.

**\*\*Conseil\*\*** : Toujours s'assurer que le contexte de sécurité **\*\*et\*\*** les données relationnelles (ex. rôles, membres) correspondent à la logique métier cible **\*avant\*** d'exécuter les actions testées.

## Annexe - Scripts generes par IA

### create project

```
package suites.unit;

import com.pfe.platform.ms_gestion.dto.request.CreateProjectRequest;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.security.SecurityUtils;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;
import com.pfe.platform.ms_gestion.service.ProjectService;

import org.mockito.InjectMocks;
import org.mockito.Mock;
import org.mockito.MockedStatic;
import org.mockito.MockitoAnnotations;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Optional;

import static org.mockito.ArgumentMatchers.any;
import static org.mockito.Mockito.*;
import static org.testng.Assert.*;

public class ProjectServiceCreateHappyTest {

    @Mock
    private ProjectRepository projectRepository;

    @Mock
    private ProjectMemberRepository projectMemberRepository;

    @Mock
    private ProjectAccessService projectAccessService;

    @InjectMocks
    private ProjectService projectService;

    private AutoCloseable mocks;
    private MockedStatic<SecurityUtils> mockedSecurity;

    @BeforeMethod
    public void setUp() {
        mocks = MockitoAnnotations.openMocks(this);
        mockedSecurity = mockStatic(SecurityUtils.class);
        mockedSecurity.when(SecurityUtils::getCurrentUserId).thenReturn(1L);
    }

    @AfterMethod
    public void tearDown() throws Exception {
        mockedSecurity.close();
        mocks.close();
    }

    @Test
    public void test_create_happy_path() {
        // Given
        CreateProjectRequest request = new CreateProjectRequest();
        request.setName("Projet Alpha");
        request.setDescription("Développement d'une nouvelle application web");

        when(projectRepository.existsByName("Projet Alpha")).thenReturn(false);
```

```

Project savedProject = new Project();
savedProject.setId(1L);
savedProject.setName("Projet Alpha");
savedProject.setDescription("Développement d'une nouvelle application web");
savedProject.setCreatedBy(1L);
when(projectRepository.save(any(Project.class))).thenReturn(savedProject);

ProjectMember savedMember = new ProjectMember();
savedMember.setId(1L);
savedMember.setProject(savedProject);
savedMember.setUserId(1L);
when(projectMemberRepository.save(any(ProjectMember.class))).thenReturn(savedMember);

// When
ProjectResponse result = projectService.create(request);

// Then
assertNotNull(result);
assertEquals(result.getName(), "Projet Alpha");
assertEquals(result.getDescription(), "Développement d'une nouvelle application web");
assertEquals(result.getCreatedBy(), Long.valueOf(1L));
verify(projectRepository).existsByName("Projet Alpha");
verify(projectRepository).save(any(Project.class));
verify(projectMemberRepository).save(any(ProjectMember.class));
}
}

```

## create envirenemnt

```

package suites.integration;

import com.pfe.platform.ms_gestion.dto.request.CreateEnvironmentRequest;
import com.pfe.platform.ms_gestion.dto.response.EnvironmentResponse;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.entity.ProjectMemberId;
import com.pfe.platform.ms_gestion.repository.CampaignRepository;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.service.EnvironmentService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextHolderImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.TestPropertySource;
import org.springframework.testng.annotations.TestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;

import static org.testng.Assert.*;

@SpringBootTest
@ActiveProfiles("test")
@Transactional
@Rollback(true)
@TestPropertySource(properties = {
    "spring.datasource.url=jdbc:h2:mem:testdb;DB_CLOSE_DELAY=-1;MODE=PostgreSQL",
    "spring.datasource.driverClassName=org.h2.Driver",

```

```
"spring.datasource.username=sa",
"spring.datasource.password=",
"spring.jpa.hibernate.ddl-auto=create-drop",
"spring.jpa.database-platform=org.hibernate.dialect.H2Dialect"
})
public class EnvironmentServiceIT extends AbstractTestNGSpringContextTests {

@Autowired private EnvironmentService environmentService;
@Autowired private ProjectRepository projectRepository;
@Autowired private ProjectMemberRepository projectMemberRepository;
@Autowired private EnvironmentRepository environmentRepository;
@Autowired private CampaignRepository campaignRepository;

private Long projectId;
private Long userId = 1L;

@BeforeMethod
public void setUp() {
// Insert prerequisite Project
Project project = new Project();
project.setName("AddProject");
project.setDescription("Test project");
project.setGitRepoUrl("https://github.com/org/webapp");
project.setCreatedBy(userId);
project = projectRepository.save(project);
projectId = project.getId();

// Insert ProjectMember to satisfy checkMembership
ProjectMember projectMember = new ProjectMember();
projectMember.setId(new ProjectMemberId(userId, projectId));
projectMember.setProject(project);
projectMemberRepository.save(projectMember);

// Set security context
SecurityContextHolder.setContext(new SecurityContextImpl(
new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
}

@AfterMethod
public void tearDown() {
SecurityContextHolder.clearContext();
}

@Test
public void test_add_happy_path() {
// Given
CreateEnvironmentRequest request = new CreateEnvironmentRequest();
request.setName("AddProject");
request.setBaseUrlWeb("https://app.example.com");
request.setBaseUrlApi("https://api.example.com");
request.setGitRepoUrl("https://github.com/org/webapp");
request.setGitBranch("develop");
request.setDatabaseType("POSTGRESQL");

// When
EnvironmentResponse result = environmentService.add(projectId, request);

// Then
assertNotNull(result);
assertNotNull(result.getId());
assertEquals(result.getName(), "AddProject");
assertEquals(result.getBaseUrlWeb(), "https://app.example.com");
assertEquals(result.getBaseUrlApi(), "https://api.example.com");
assertEquals(result.getGitRepoUrl(), "https://github.com/org/webapp");
assertEquals(result.getGitBranch(), "develop");
assertEquals(result.getDatabaseType(), "POSTGRESQL");

// Verify persistence in DB
Environment savedEnv = environmentRepository.findById(result.getId()).orElse(null);
assertNotNull(savedEnv);
```

```
assertEquals(savedEnv.getName(), "AddProject");
assertEquals(savedEnv.getProject().getId(), projectId);
}
}
```

## List environnement

```
package suites.integration;

import org.springframework.test.context.TestPropertySource;

import com.pfe.platform.ms_gestion.MsGestionApplication;
import com.pfe.platform.ms_gestion.dto.request.CreateEnvironmentRequest;
import com.pfe.platform.ms_gestion.dto.response.EnvironmentResponse;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.service.EnvironmentService;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.testng.AbstractTestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;

import static org.testng.Assert.*;

@SpringBootTest(classes = MsGestionApplication.class)
@ActiveProfiles("test")
@Transactional
@Rollback(true)
public class EnvironmentServiceIT extends AbstractTestNGSpringContextTests {

    @Autowired private EnvironmentService environmentService;
    @Autowired private EnvironmentRepository environmentRepository;
    @Autowired private ProjectRepository projectRepository;

    private Long projectId;
    private Long userId;

    @BeforeMethod
    public void setUp() {
        userId = 1L;

        Project project = new Project();
        project.setName("project-alpha");
        project.setDescription("Test project");
        project.setGitRepoUrl("git@github.com:example/project-alpha.git");
        project.setCreatedBy(userId);
        project = projectRepository.save(project);
        projectId = project.getId();

        SecurityContextHolder.setContext(new SecurityContextImpl(
            new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
    }

    @AfterMethod
```

```
public void tearDown() {
    SecurityContextHolder.clearContext();
}

@Test
public void test_list_happy_path() {
    // Given
    CreateEnvironmentRequest request = new CreateEnvironmentRequest();
    request.setName("project-alpha");
    request.setBaseUrlWeb("https://app.example.com");
    request.setBaseUrlApi("https://api.example.com");
    request.setGitRepoUrl("git@github.com:example/project-alpha.git");
    request.setGitBranch("develop");
    request.setDatabaseType("postgres");

    environmentService.add(projectId, request);

    // When
    java.util.List<EnvironmentResponse> result = environmentService.list(projectId);

    // Then
    assertNotNull(result);
    assertFalse(result.isEmpty());
    assertEquals(result.size(), 1);

    EnvironmentResponse env = result.get(0);
    assertNotNull(env);
    assertEquals(env.getName(), "project-alpha");
    assertEquals(env.getBaseUrlWeb(), "https://app.example.com");
    assertEquals(env.getBaseUrlApi(), "https://api.example.com");
    assertEquals(env.getGitRepoUrl(), "git@github.com:example/project-alpha.git");
    assertEquals(env.getGitBranch(), "develop");
    assertEquals(env.getDatabaseType(), "postgres");
}
}
```